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Dear Professor Hans Irschik

We are deeply grateful to you for having carefully revised our work yourself.

We believe all the points evoked were resolved.

In particular, the references issue has been carefully addressed, and now they appear in alphabetical order with numbering. Also, the question marks have been removed. We are regretful about that.

A few other points, concerning some plots, labels and equations, were clarified.

Also, a few phrases have been changed, for a better understanding of the message of the paper, we believe.

Those points appear in red, in the new version of the manuscript.

We thank you again, Professor, and hope this new submission is now adequate for publishing in the prestigious Acta Mechanica.

Best wishes

Lucas Franceschini

The Mechanics of Variable Mass Systems applied to the Added Mass Concept of a Moving Cylinder in Water

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Abstract

This paper constructs a physics-based surrogate mathematical model for the motion of a cylinder surrounded by a fluid under a distinct point of view: the mechanics of variable mass systems. Particularly, the concept of added mass is addressed. First, we introduce a 1 DoF problem, consisting of an equivalent virtual particle, free to oscillate in one direction, whose mass is let to depend on its position and velocity. The resulting kinetic energy models that of the whole system, solid and surrounding fluid. A general formulation for the equation of motion is then proposed, by applying the Extended Lagrange Equations for variable mass systems, from which the surrogate model is derived. We take as first case study the classic Vortex-Induced Vibration (VIV) phenomenon of a cylinder mounted on an elastic base. Then, we assume the added mass as a polynomial function on position and velocity. The coefficients of this polynomial expansion are estimated by regression, where we minimize the residual between the model's response and external data, herein coming from Computational Fluid Dynamics (CFD) simulations. The results are rewarding and the well known behavior of the added mass as a function of the reduced velocity observed in the technical literature, from experiments and from CFD simulations, is consistently recovered.

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1 Introduction

Problems where variable-mass systems formulations are employed have been extensively studied in the past. A classical example would be the so-called "rocket problem", which burns and expels fuel over time, changing the mass within a control-volume corresponding to the carcass of the rocket. Another example, would be the classic falling chain problem ([5]), in which the mass of the suspended part varies with the tip position, that may be taken as a generalized coordinate. More recently, the Euler-Lagrange equation of such variable-mass systems has been derived (see [26]), allowing for a systematic way to establish the equations of motion, in cases where the mass depends not only explicitly on time, but also on position and velocity. For a review of such equations, written in a few different forms see [25], chapter 8. Many applications in classical mechanics have been studied through this lens (in addition to the falling chain, we may also point out the impact of solid body on a liquid surface, as in [26] or [4, 3] for a more detailed description; the oscillating water column, see [30, 7] and [25], chapter 8); or even the dynamics of vertically collapsing buildings (see [27]), all of which had the mass dependent on some generalized position only. Although the extended Lagrange equation in [26] has been derived for systems with mass explicitly dependent on time, position and velocity, the latter dependence has never been addressed in practical problems. Indeed, it is quite hard to conceive a mechanical system with mass depending explicitly on its velocity. However, an attempt to do so could be to use mass in a virtual manner, such as the hydrodynamical added mass concept in the context of viscous flows.

The general concept of added mass (see, for example, [21]) aims at modeling the component of the force (generally reactive) exerted by a fluid on a moving solid, which is in anti-phase with its acceleration. In addition, this added mass is not dependent on the velocity of the solid, but just on its position, if in the proximity of another solid body or a free surface. We remark that, as the flow is ideal, the added mass can be seen as a geometrical concept, particular to the solid's shape and immersed volume (or drift volume). This point of view was explored in the work of [8], and corresponds to the so-called Darwin's drift. Although firstly introduced (and fully justifiable) in the context of potential flow theory, the added mass concept can be extended empirically to model more complex scenarios in viscous flows, such as Vortex Induced Vibrations (VIV). In VIV, we typically witness the lock-in phenomenon, where the frequency of Vortex-Shedding (f_{VS} , or the external excitation frequency, from the point of view of the solid) matches the solid's natural frequency (f_n) and stays related to it when f_{VS} is increased, when augmenting the incoming flow velocity. This shift in frequency can be thus modeled as an added mass effect (see [29]), dependent on the frequency (or reduced velocity). This added mass is positive for frequencies f_{VS} below f_n (as f_{VS} dominates), zero for $f_{VS} \approx f_n$, negligible within the lock-in range, and negative for larger f_{VS} (see [10]). It turns out that this behavior is much more general, where any system, being forced with a well-defined-frequency forcing (see [6]) exhibit such behavior as well. More recently, a time-varying added mass (based on Darwin's drift, which allowed to experimentally compute it), the so-called energized-mass, was introduced by [14] and allows for the modeling of viscous effects in time-evolving boundary layers ([9]). In this framework, the mass of the equivalent mechanical model

varies with time, and the Euler-Lagrange equations in [26] have been used in the derivation of the system's equations. However, although the energized-mass model can well reproduce the data, it does not provide a description based on a fully autonomous (whose mathematical structure is time-independent) mechanical system. To do so, its time-dependency would need to be replaced with the dependency on the system's state (position and velocity). This dependency would encode not only how the vortical pattern in the flow would force the moving solid, but would also have some memory effect (due to dependency on velocity), modeling viscous effects and boundary layer lags.

The present work can be divided in two main parts. First a system's dynamical equations are derived based on the Euler-Lagrange formalism presented in [26], where the mass is allowed to be a general function of position and velocity. In this formulation, the mass that is added/lost by the particle has a certain velocity, and a discussion around how this quantity is modeled will be provided. Then, in a second part, the mass is allowed to be a polynomial function of position and velocity, whose coefficients are determined via regression on data coming from numerical simulations (Computational Fluid Dynamics, CFD). This step is similar to recent techniques such as SINDy (see [2] for its original proposition and [18] in the context of VIV). We should also cite a more recent work from [15], who employed an algorithm to determine the system's inertia, also function of the state, based on the Hilbert Transform ([23]), which, in a first step the frequency is determined as function of time, and further correlated to the state equation which also has a state-dependent natural frequency.

2 The variable mass system approach and the 1 DoF problem

First of all, let us introduce the general variable-mass model, which is intended to describe, as a surrogate model, the whole Vortex-Induced Vibration (VIV) problem by the virtue of the mass being explicitly dependent on the position, velocity and time [26]. The starting point will be the extended Lagrange Equation valid this type of systems:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j^a + Q_j^t + Q_j^y + Q_j^i \quad (1)$$

with

$$\begin{aligned}
Q_j^a &= \sum_i \mathbf{f}_i \cdot \frac{\partial P_i}{\partial q_j} \\
Q_j^t &= \sum_i \dot{m}_i \mathbf{w}_i \cdot \frac{\partial P_i}{\partial q_j} \\
Q_j^q &= - \sum_i \left(\frac{1}{2} \frac{\partial m_i}{\partial q_j} |\mathbf{v}_i|^2 \right) \\
Q_j^{\dot{q}} &= \frac{d}{dt} \sum_i \left(\frac{1}{2} \frac{\partial m_i}{\partial \dot{q}_j} |\mathbf{v}_i|^2 \right)
\end{aligned} \tag{2}$$

In Equation 2, $m = m(t, q_j, \dot{q}_j)$ is the mass of the j -th particle, explicitly depending on time, generalized coordinates and their time derivatives; $\mathbf{f}_i$ is an external force applied to the particle; $\mathbf{v}_i$ is its velocity and $\mathbf{w}_i$ is the velocity of the lost (gained) mass from (to) the particle, both measured w.r.t. the same inertial and cartesian frame of reference. Notice that the three latter terms in Equation 2 are respectively functions of derivatives of the particle mass w.r.t. time, position, and velocity. For the case where $m = m(t, q_j)$, those equations are also presented in other forms, see [25], chapter 8.

Let us now use this formalism to model the behavior of a rigid body, immersed in water and allowed to move along a single Degree of Freedom (DoF) denoted y . For this, we should first write the kinetic energy of the whole system, composed of the rigid body and the fluid around it. For that, we will let the kinetic energy to be written as the one of a single equivalent particle:

$$T = \frac{1}{2} m(t, y, \dot{y}) \dot{y}^2 \tag{3}$$

Here, the mass of the equivalent particle $m(t, y, \dot{y})$ will be dependent on position, velocity and time. In general, this mass will be given by the sum of the (constant) solid's mass, m_c and an added mass contribution $m_a = m_a(t, y, \dot{y})$:

$$m(t, y, \dot{y}) = m_c + m_a(t, y, \dot{y}) \tag{4}$$

Then, the corresponding extended Lagrange Equation is written,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{y}} \right) - \frac{\partial T}{\partial y} = Q^a + Q^t + Q^y + Q^{\dot{y}} \tag{5}$$

with

$$\begin{aligned}
Q^a &= F(t, y, \dot{y}) = F_m + R \\
Q^t &= \dot{m} w \\
Q^y &= - \frac{1}{2} \frac{\partial m}{\partial y} \dot{y}^2 \\
Q^{\dot{y}} &= \frac{d}{dt} \left(\frac{1}{2} \frac{\partial m}{\partial \dot{y}} \dot{y}^2 \right)
\end{aligned} \tag{6}$$

In the term $F(t, y, \dot{y})$ we include all the other forces acting on the cylinder, which are not of inertial nature, written as a sum of F_m and R . The F_m term consists in the modeled forces, which are supposed to affect the dynamics, such as elastic or damping ones. In the term R we include the residual forces that could not be taken into account in the model F_m .

Here, w is the velocity of the lost (gained) mass w.r.t. the same inertial frame of reference used for measuring y (see [26]). As a first ansatz, this velocity might be taken as null, $w = 0$, what is known as the Levi-Civita case in the mechanics of variable mass systems [26]. An alternative for that, would be to take $w = \dot{y}$, the same velocity of the studied particle, set equal to the solid body's one. Such an alternative will be discussed further in this paper. For simplicity, we let the velocity of the gained or lost mass be a linear function of the equivalent particle's velocity, i.e.,

$$w(t) = \lambda \dot{y}(t) \quad (7)$$

where λ is taken as a constant. According to λ , we may have different cases:

- $\lambda = 0 \rightarrow w(t) = 0$: the particle acquires/loses mass at zero velocity. This is commonly known as the Levi-Civita case (see [26])
- $\lambda = 1 \rightarrow w(t) = \dot{y}(t)$: the particle acquires/loses mass at the same velocity it moves
- $\lambda \neq 0$ or $\lambda \neq 1$: the particle acquires/loses mass at some other velocity

We observe that this is a (not unique) choice that simplifies the problem. Other models could be considered, including additional transfer functions between the input ($\dot{y}$) and the output (w). This could be investigated in future work.

With this settings, the terms in the extended Euler-Lagrange equation are given by:

$$\frac{\partial T}{\partial y} = \frac{1}{2} \left(\frac{\partial m}{\partial y} \right) (\dot{y})^2, \quad \frac{\partial T}{\partial \dot{y}} = \frac{1}{2} \frac{\partial m}{\partial \dot{y}} (\dot{y})^2 + m \dot{y} \quad (8)$$

and

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{y}} \right) &= \frac{d}{dt} \left(\frac{1}{2} \frac{\partial m}{\partial \dot{y}} (\dot{y})^2 + m \dot{y} \right) \\ &= \frac{d}{dt} \left(\frac{1}{2} \frac{\partial m}{\partial \dot{y}} (\dot{y})^2 \right) + \left(\frac{\partial m}{\partial t} + \frac{\partial m}{\partial y} \dot{y} + \frac{\partial m}{\partial \dot{y}} \ddot{y} \right) \dot{y} + m \ddot{y} \end{aligned} \quad (9)$$

We can see that, a few terms involving derivatives of the mass w.r.t. the state cancel out with Q^y and $Q^{\dot{y}}$, leading to:

$$\begin{aligned} \left(m_c + m_a + \frac{\partial m_a}{\partial \dot{y}} (1 - \lambda) \dot{y} \right) \ddot{y} + \left(\frac{\partial m_a}{\partial y} \right) \dot{y}^2 (1 - \lambda) \\ + \left(\frac{\partial m_a}{\partial t} \right) \dot{y} (1 - \lambda) = F(t, y, \dot{y}) \end{aligned} \quad (10)$$

2.1 The equations of motion for a cylinder under VIV

So far, the equations of motion have been given in a general manner, for a single degree of freedom, variable mass system. We further particularize the model, taking a cylinder attached to a linear spring-damper system, of known constants k and c . Also, in order to model extra components of the fluid force on the solid, which were not necessarily captured by the (variable) added mass, we further add some typical non-linear viscous damping, whose force is given by $\tilde{\chi}|\dot{y}|\dot{y}$. The generalized force may be then written,

$$F_m(t, y, \dot{y}) = -c\dot{y} - ky - \tilde{\chi}|\dot{y}|\dot{y} \quad (11)$$

The surrogate model is finally given by:

$$\begin{aligned} \left(m_c + m_a + \frac{\partial m_a}{\partial \dot{y}}(1 - \lambda)\dot{y} \right) \ddot{y} + \left(\frac{\partial m_a}{\partial y} \right) \dot{y}^2(1 - \lambda) \\ + (c + \tilde{\chi}|\dot{y}|)\dot{y} + ky + \frac{\partial m_a}{\partial t}(1 - \lambda)\dot{y} = R \end{aligned} \quad (12)$$

Notice that $\lambda = 1$ leads to the simplest form for this surrogate model, making null all the terms coming from $\dot{m}_a = dm_a/dt$. This is going to be further explored in the paper, not only as the simplest form but also as a natural choice.

2.2 Non-dimensional model

Before going any further, we rewrite here the dimensionless form of the surrogate model, which will be convenient for further generic development. To do so, we give ourselves a typical length scale D (taken here as the cylinder's diameter) and a typical velocity scale U (to be precised later) and a resulting time-scale D/U , such that we can define:

$$\tau = \frac{t}{D/U}, \quad y = \frac{y}{D}, \quad \dot{y} \equiv \frac{d}{d\tau}y = \frac{\dot{y}}{U}, \quad \ddot{y} \equiv \frac{d^2}{d\tau^2}y = \frac{\ddot{y}D}{U^2} \quad (13)$$

where here we abuse the notation $(\dot{\cdot})$ for derivatives with respect to both t and τ . We may also define:

$$a \equiv \frac{m_a}{m_c} = a(\tau, y, \dot{y}), \quad (14)$$

to be the dimensionless added mass, function of $(\tau, y, \dot{y})$. Alternatively, as it is more usual in some studies, we may also make the added mass m_a dimensionless with the displaced volume mass, $m_d = \rho\pi D^2/4$:

$$C_a = \frac{m_a}{m_d} = am^*, \quad \text{with} \quad m^* = \frac{m_c}{m_d}. \quad (15)$$

where ρ is the density of the surrounding fluid. Replacing those identities in equation 12, we get:

$$\begin{aligned}
& \left(1 + a + \frac{\partial a}{\partial \dot{y}}(1 - \lambda)\dot{y}\right) \ddot{y} + \left(\frac{\partial a}{\partial y}\right) (\dot{y})^2(1 - \lambda) \\
& + \left(4\pi \frac{\zeta}{U_r} + \chi|\dot{y}|\right) \dot{y} + \left(\frac{2\pi}{U_r}\right)^2 y + \frac{\partial a}{\partial \tau}(1 - \lambda)\dot{y} = \frac{R}{m_c U^2/D} \equiv r
\end{aligned} \tag{16}$$

with:

$$\zeta = \frac{1}{2} \frac{c}{\sqrt{k m_c}}, \quad f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m_c}}, \quad U_r = \frac{U}{D f_n}, \quad \chi = \frac{\tilde{\chi}}{m_c D} \tag{17}$$

the structural damping ratio ζ , the solid's natural frequency f_n (without the constant added mass correction, as it will be found by the data-driven regression process afterwards), the reduced-velocity U_r (based on f_n , as in [24]) and the dimensionless nonlinear damping coefficient χ .

2.3 Model for the variable mass, $m_a = m_a(t, y, \dot{y})$ (or $a = a(\tau, y, \dot{y})$)

At this point, the added mass of the cylinder has been modeled, in all its generality, as function of time t , position y and velocity $\dot{y}$, $m_a = m(t, y, \dot{y})$, or, in its dimensionless form $a = a(\tau, y, \dot{y})$. The explicit time-dependency of the mass with time has been recently addressed by the introduction of the so-called energized mass concept [13]. Here, instead of letting it be explicitly function of time, we only let it be function of the position and velocity $(y, \dot{y})$, its time-dependency being implicitly given by the time variation of those variables. This latter view is more aligned with the philosophy of providing an autonomous description of the system.

We further suppose the added mass to be dependent on position and velocity as a polynomial:

$$a = a(y, \dot{y}) = \sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i,j} y^i \dot{y}^j \tag{18}$$

This is commonly done in system identification literature (see, for example [1, 2, 22]) and is convenient when no better or more obvious functional basis is at hand. However, each polynomial term has a physical interpretation. For instance, the first one, $\gamma_{0,0}$ is a time-invariant term and provides the classical "constant" added mass term, first introduced for potential flows and latter extended to the VIV phenomenon as function of the reduced velocity (see [28]). The terms involving the position-dependency only, i.e. $\gamma_{1,0}y + \gamma_{2,0}y^2 + \dots$ can also be interpreted as the portion of the added mass that models purely space-dependent masses, as in the case where nearby solid walls or free-surfaces are present. The remaining terms are responsible for modeling velocity-dependent masses, and thus for memory effects. With this model, the derivatives of the added mass w.r.t. y and $\dot{y}$ are given by:

$$\begin{aligned}
\frac{\partial a}{\partial y} &= \sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i+1,j} (i+1) y^i \dot{y}^j \\
\frac{\partial a}{\partial \dot{y}} &= \sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i,j+1} (j+1) y^i \dot{y}^j
\end{aligned} \tag{19}$$

We remark that even if the expression for the added mass a is polynomial in y and $\dot{y}$, the whole dynamics governing the state $(y, \dot{y})$ is not polynomial and has a rather complex structure. Indeed, the representation in state-space of this dynamical system would lead to a fractional function, whose numerator not only contains polynomials, but also a function of the type $|\dot{y}| \dot{y}$, if $\chi \neq 0$. In terms of model regression approaches, this does not correspond to a typical early versions of SINDy [2], although closer to more recent versions of it such as py-SINDy [19]. Notice that, in our case, this representation comes naturally from the variable-mass Lagrangian formalism [26].

We also remark that, from a physical point of view, the only known parameters in the model are the mass of the cylinder, m_c , the linear damping c and the stiffness, k , leading to known values for ζ , f_n and U_r . All the other parameters, namely $\gamma_{i,j}$ and χ are "virtual" and will be determined by a regression step, as described in the next section. The parameter λ , which determines how the velocity of acquired mass will be related to the cylinder's velocity, will be assessed.

2.4 Reconstruction of added mass with regression of data

This section is devoted to the numerical procedure for determining the coefficients $\gamma_{i,j}, \chi$. We suppose we have at hand a time-series data for $y(t)$. In our case, this data will come from Computational Fluid Dynamics (CFD) simulations. However, experimental data could be used as well. With this data we proceed to compute its derivatives, up to second order $(\dot{y}, \ddot{y})(t)$, for instance using second-order finite difference scheme. The coefficients $\gamma_{i,j}$ of each monomial and χ will be found by performing a least-square regression, where we minimize the cost functional:

$$J(\gamma_{i,j}, \chi) = \frac{1}{2} \|r(\gamma_{i,j}, \chi)\|_2^2 + \frac{\sigma}{2} \left(\sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i,j}^2 + \chi^2 \right) \tag{20}$$

being the second term a regularization component, penalizing equally the square of each coefficient $\gamma_{i,j}$ and χ . We emphasize here that, due to the adimensionalization performed before, all those terms in the equation are also dimensionless and can be compared with each other.

3 Application to Vortex-Induced Vibration Case

3.1 The numerical approach to the fluid-structure problem

As a starting point for our analysis, we consider the fluid-structure interaction phenomenon described before, but regarded as solution of the fluid flow equations, coupled with the solid dynamics. In what follows, we choose to directly present this problem in dimensionless form, following the same procedure presented for the surrogate model, where the mass, the length and velocity used, m_c, D, U are still the cylinder's mass and diameter and U will be chosen here as the velocity of the incoming flow (uniformly distributed far from the cylinder), respectively. With that, the cylinder's displacement y , will be modeled by:

$$\ddot{y} + 4\pi \frac{\zeta}{U_r} \dot{y} + \left(\frac{2\pi}{U_r} \right)^2 y = g_y \equiv \frac{4}{\pi} \frac{1}{m^*} f_y \quad (21)$$

where f_y is the force from the fluid flow on the direction of motion y , and can be computed as an integral over the solid wall Γ_s of the (dimensionless) stress tensor:

$$\sigma = pI - Re^{-1}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad (22)$$

where $Re = \rho UD/\mu$ is the Reynolds number, based on the typical fluid velocity U , the cylinder's diameter D , the fluid density ρ and viscosity μ . This tensor is function of the pressure field $p = p(\mathbf{x})$ and of the gradients of the velocity field $\mathbf{u} = (u_1, u_2) = \mathbf{u}(\mathbf{x})$ (also dimensionless), along the normal direction $\mathbf{n}$ and projected onto the versor aligned with y , $\mathbf{e}_2$ as:

$$f_y(\mathbf{u}, p) = \int_{\Gamma_s} \mathbf{e}_2 \cdot (p \cdot I - \mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T)) \cdot \mathbf{n} dS \quad (23)$$

Here, $\mathbf{x} = (x_1, x_2)$ is a point in the space where the fluid-flow is defined, and is supposed two-dimensional, and $\nabla = (\partial_{x_1}, \partial_{x_2})$. The coordinates x_1 (with unit vector $\mathbf{e}_1$) and x_2 (with unit vector $\mathbf{e}_2$) correspond to the streamwise and crosswise directions.

The fluid motion, driven by an imposed far-field velocity aligned with $\mathbf{e}_1$ and normalized, $\mathbf{u}_\infty = 1\mathbf{e}_1$, is governed by the Navier-Stokes equations:

$$\partial_\tau \mathbf{u} + (\mathbf{u} - \mathbf{u}_m) \cdot \nabla \mathbf{u} + \nabla p = Re^{-1} \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0 \quad (24)$$

Here we make a choice of writing the Navier-Stokes equations in the Arbitrary Lagrangian Eulerian (ALE) form (see for example [17]), as it will be the choice for our numerical implementation when taking the mesh-velocity, $\mathbf{u}_m$, as the cross-wise velocity of the cylinder, $\mathbf{u}_m = \mathbf{e}_2 \dot{y}$.

The Reynolds number will be kept constant and equal to $Re = 100$ in all our simulations (as in [24]). The mass-ratio will also be kept constant and equal to $m^* \approx 6.4$ [24]. Also, no structural damping will be considered ($\zeta = 0$). The reduced velocity U_r will be varied within a range of values of $U_r \in [4, 9]$. We remark that by taking both the Reynolds and the reduced mass constants, we fix both the vortex-shedding frequency (characterized by a Strouhal number of $St = f_{VS} D / U_\infty \approx 0.18$, which is computed with a fix-cylinder CFD simulation) and the mass/diameter of the cylinder. For this reason, if we were to reproduce those settings experimentally, we would have to

vary the stiffness of the spring on which the cylinder is mounted (instead of increasing the flow velocity, which is more usual in practice [12]).

In order to study this problem, we simulate it using the finite-element code FreeFEM++ [16] for the spatial discretization of the fluid problem. A second-order backward finite difference for the time-derivative of the fluid problem [20] and a centered finite-difference in time is used for the solid equations.

3.2 Numerical Simulations data

The results of several time-stepping simulations, started with initial conditions corresponding to the steady-state (base-flow) solution added with a small perturbation along the unstable mode, coming from spectral analysis of the linearized Navier-Stokes equations (corresponding to the vortex-shedding mode multiplied by a small amplitude) can be appreciated in Figure 1 for $U_r = 4, 4.5, 5$ and 8 . We can see that we recover the usual behavior where for small reduced velocities, the amplitude of motion is small and it is caused by the forcing from the fluid, which oscillates at the vortex-shedding (VS) frequency, typically characterized by a Strouhal number $St = f_{VS}D/U_\infty \approx 0.18$. For $U_r \geq 5$, the cylinder's natural frequency matches the VS one, and a resonance occurs, increasing the amplitude of motion. This is the beginning of the lock-in region. If we further increase U_r , we still witness a large amplitude of motion and the frequency of that motion is locked to the cylinder's natural frequency, even if the VS one is not matching exactly that frequency. We see that the dimensionless frequency $f^* \equiv f/f_n = f(U_\infty/D)/U_r$ becomes unity in this region. For much larger reduced velocities, near $U_r = 8$, we begin to witness the end of this phenomenon, where the amplitude of the motion becomes again smaller and the frequencies in the signal slowly start to transition to the fluid one. A summary of this description is given in Figure 2 showing the amplitude of motion and the Power Spectral Density (PSD) of the signal (always computed for $\tau > 100$, the fully-converged portion) for each reduced velocity considered here. We remark that, although this results are very close from the one reported by [24], obtained in a simplified, low-Reynolds number, two-dimensional setup, it is qualitatively different from more usual lock-in diagrams (for example, [11, 12]), obtained experimentally, when higher Reynolds numbers are involved, three-dimensional effects and elasticity of the cylinder has to be taken into account. This simplification is employed for the sake of the methodology presentation and validation.

3.3 A first regression

In this section, we present some results in a first regression test, corresponding to polynomial degree of 2 in both y and $\dot{y}$, together with the regression in $\chi\dot{y}|\dot{y}|$. The coefficients $\gamma_{i,j}$ for that regression are given in Figure 3, for three different regularization parameters $\sigma = 0.01, 0.1$ and 0.4 , and three different λ , which are $\lambda = 0$ (the mass is acquired at zero velocity, Levi-Civita case), $\lambda = 1$ (the mass is acquired at the same velocity as the particle) and an intermediate value of $\lambda = 0.5$. We can see that the first coefficient, $\gamma_{0,0}$ is nearly the same for all regularization parameters and for all acquisition velocity parameters. This means that, regardless of the physics of the

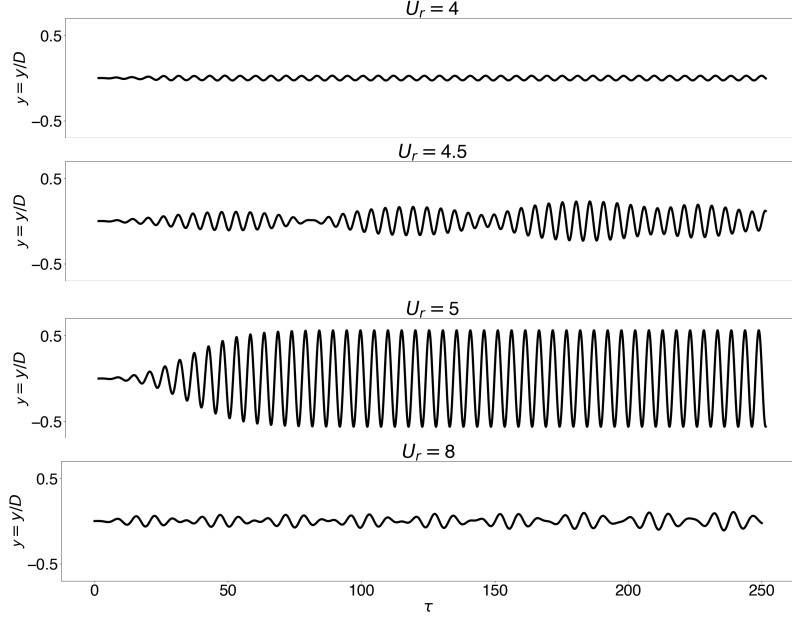


Fig. 1 Signal from cross-wise ($y(\tau)$) of the cylinder for $U_r = 4, 4.5, 5$ and 8 , determined from CFD simulations.

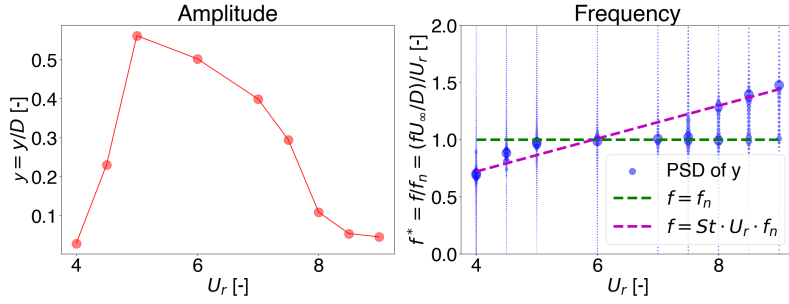


Fig. 2 Amplitude and Power Spectral Density (PSD, whose energy is proportional to the diameter of the circles) for displacement signal y for several reduced velocities. The typical lock-in scenario is recovered where higher-amplitudes are obtained for a reduced-velocity interval starting from $U_r \approx 5$, from which its frequency stops being correlated with the vortex-shedding one (characterized by a Strouhal number of $St \approx 0.18$) and correlates with the natural frequency of the solid $f = f_n$.

added mass process and of the finer details of each regression, the constant term persists. Furthermore, we can interpret it as being the added mass coefficient, having a higher-value before the lock-in (where it has to adjust the system's natural frequency to match the one from the vortex-shedding phenomenon) a zero-value within the lock-in (where the cylinder's natural frequency is already correct) and a negative value for post lock-in region (where again vortex-shedding dominates). The behavior for this coefficient had been reported for example in [6, 10, 33] as experimental results and, for instance, in [34] by numerical simulations.

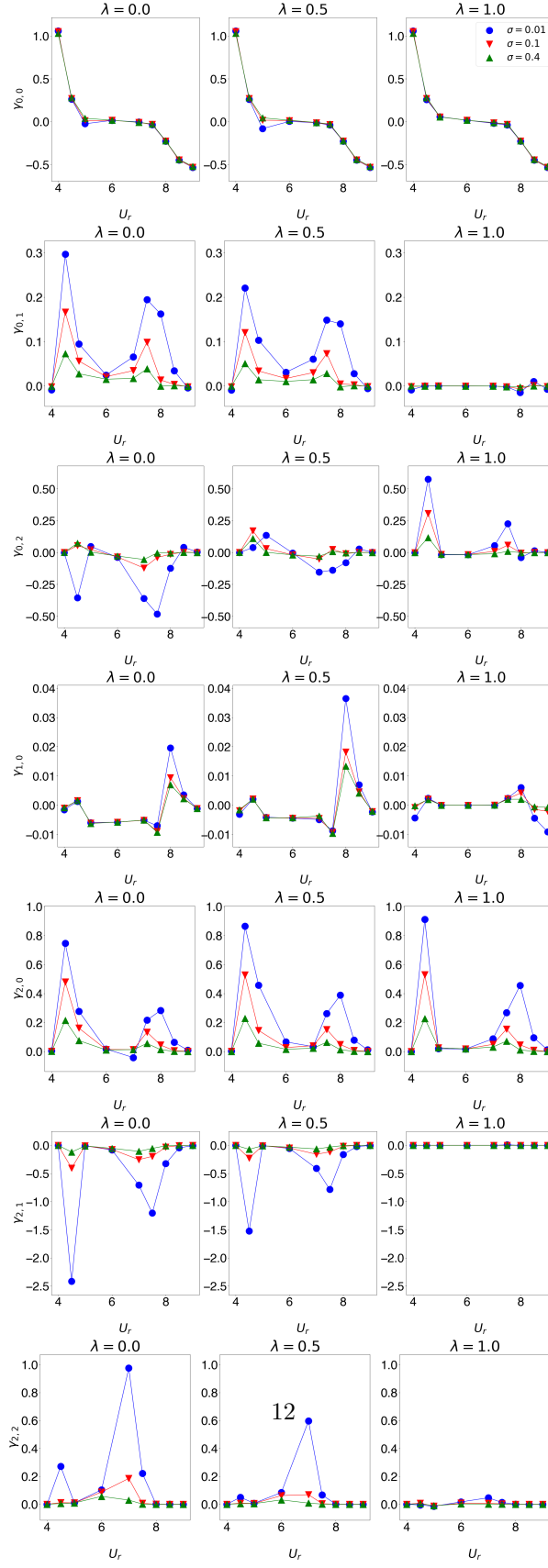


Fig. 3 Regression coefficients for the added mass, $\gamma_{i,j}$, as function of the reduced velocity U_r .

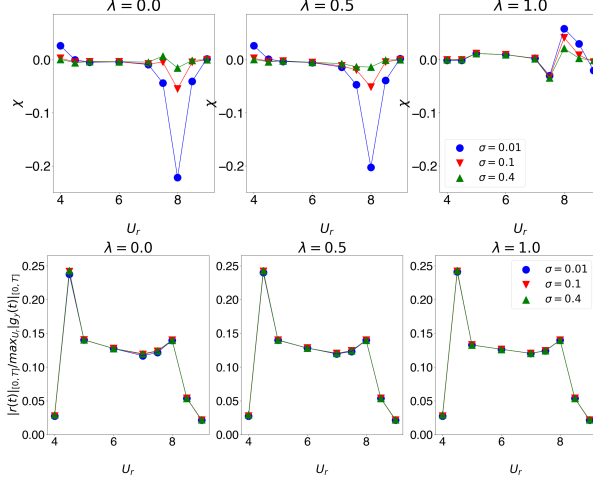


Fig. 4 Regression coefficient χ and residual error (given as $|r(\tau)|_{[0,T]} \equiv \int_0^T |r(\tau)|^2 dt$, normalized by $\max_{U_r} |g_y(\tau)|_{[0,T]}$), both function of the reduced velocity U_r . Here, g_y is defined in Equation 21.

On top of this first adjustment of the system's natural frequency, we can observe the other polynomial coefficients for the model, $\gamma_{i,j}$. We can see that their behavior is typically active for reduced velocities where both solid and fluid frequencies are present (for example $U_r \approx 4.5$ and $U_r \approx 8$), showing that the surrogate model is trying to capture this more complex behavior. Interestingly, for the Levi-Civita case, all the model's coefficients exhibit a more intricate distribution at those regions, presenting higher values for $\gamma_{i,j}$. We can interpret this as coming from the fact that for $\lambda = 0$ (as opposed to $\lambda = 1$), the model has a more complex structure, exhibiting more nontrivial terms in its dynamics (Equation 20), to be adjusted by the regression process. However, for $\lambda = 1$, the model becomes much simpler. This does not seem to be an issue for the regression error, given in Figure 4, which stays almost the same for all values of λ considered. As a matter of fact, $\lambda = 1$ seems a natural choice, considering the physical nature of the surrogate model, such that no 'continuous impacting force' is implied, as would be the case for $\lambda \neq 1$ (for instance, the Levi-Civita hypothesis, $\lambda = 0$), that seems unnatural. Furthermore, we can see that the error is practically the same for all regularization parameters σ as well. For this reason, in the following, we will take a closer look at the case $\lambda = 1$ and $\sigma = 0.4$, as they show to be the simplest ones.

For this case, we can see that the only active terms, besides $\gamma_{0,0}$ in the regression are $\gamma_{1,0}$, $\gamma_{2,0}$ and $\gamma_{0,2}$. The terms $\gamma_{1,0}$ and $\gamma_{2,0}$ represent a linear and a quadratic dependency on the position y . Having such dependency may seem counter-intuitive as it does not respect the y -symmetry of the VIV problem (i.e. if $(y(t), (u_1(x_1, x_2), u_2(x_1, x_2)), p(x_1, x_2))$ is a solution, so is $(-y(t), (u_1(x_1, -x_2), -u_2(x_1, -x_2)), p(x_1, -x_2))$). Indeed, although the mass of the system has a quadratic dependency on y (which is symmetric w.r.t. negative/positive values of y), it also has a linear dependency on y , implying that different masses would be obtained for negative or positive values of y . From the point of view of potential flow (for which the added mass concept has first been introduced), the added mass should

only depend on the position if, for example, solid walls near the cylinder are present, which would make the flow's streamlines to deform and affect differently the cylinder according to its position. However, we must have in mind that, according to the position of the cylinder, $y(\tau)$ (differently from the potential-flow case), the flowfield will exhibit a pattern corresponding to a given phase of the vortex-shedding motion. This pattern will affect the solid body in a different manner, according to the Biot-Savart law, translating the vortices in the fluidflow into a force, whose effect is modeled by the added mass. The term $\gamma_{0,2}$ represents a quadratic dependency in the velocity y for the added mass m_a , which is positive in all scenarios. This means that, if the cylinder is moving faster, its effective mass is higher. We remark that, for $\lambda = 1$ and $\sigma = 0.4$, this effect will mostly be present for $U_r \approx 4.5$ and $U_r \approx 8$, transitional regimes which present a more complex behavior.

Lastly, we should test the surrogate model and compare it with the actual signal coming from the original CFD results. The model has been simulated using an Ordinary Differential Equation (ODE) from "scipy". The initial condition for those simulations correspond to the actual state of the cylinder at $\tau = 100$ and are integrated on the same time grid up to $\tau = 250$. The results are provided in Figure 5.

As pointed out previously, due to an increased complexity of the signal, for $U_r = 4$ and $U_r = 5$, the model, for the three regularization parameters, perform very well. For the case $U_r = 4.5$, even with a complex added mass dependency on $(y, \dot{y})$, the model behaves essentially as a linear oscillator, not being able to fully exhibit the modulation presented before. Also, the model for $U_r = 8$ is not fully able to well represent the dynamics of the two distinct frequencies, as it seems only one is effectively present in the signal. This is also verified by Figure 6, where only one frequency is recovered by the PSD, whereas two distinct values are present in the original signal in Figure 2. Despite those deficiencies, we can see that the overall amplitude produced by the model captures well the actual one, showing that the model (although simple) can be used as first-order approximation of the dynamics.

3.4 A finer regression

The question that arises now is whether taking a higher polynomial order for the model can mitigate some of the deficiencies of the regressed model. To do so, we take now a fourth-order polynomial in both y and $\dot{y}$ variables. Instead of showing the coefficients, it is clearer and more concise to only show the error committed by the model, as presented in Figure 7. We can see that by taking a higher-order model can only slightly mitigate some of the error especially for the less regularized case, $\sigma = 0.01$ and also for $\lambda = 0$. This is due to a richer structure of the surrogate model, as discussed before. Furthermore, this slight reduction in the error occurs in the lock-in region, when the signal is rather simple, having essentially the natural frequency of the structure. This means that the essential of the dynamics has been captured with the previous model.

For the present model, also for the most regularized case, $\sigma = 0.4$, we present the added mass, for a few values of reduced velocity, as function of y and $\dot{y}$ in Figure 8. We can see that, for $U_r = 4$, $a = m_a/m_c$ is almost constant and equal to the added mass coefficient of nearly $a \approx 1$. For $U_r = 8$, a is also relatively constant, having $\gamma_{0,0}$ as its main contribution, but negative this time. For $U_r = 4.5$ we can see a slightly stronger

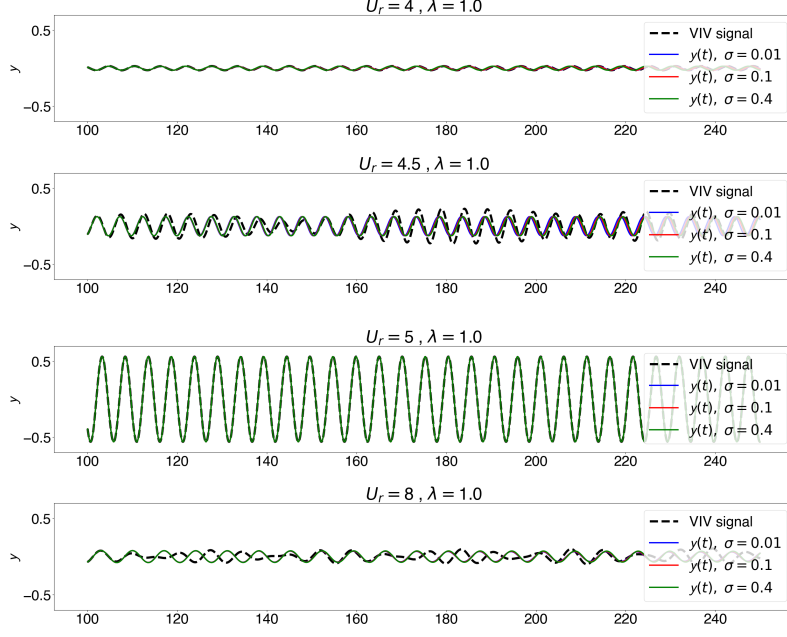


Fig. 5 Signal from cross-wise displacement (y) for the model's outputs (in colors; given by equation 20) and the comparison with original, VIV signal, in black dashed lines, for $U_r = 4, 4.5, 5$ and 8 . The lines are almost superposed, as in Figures 6 and 9.

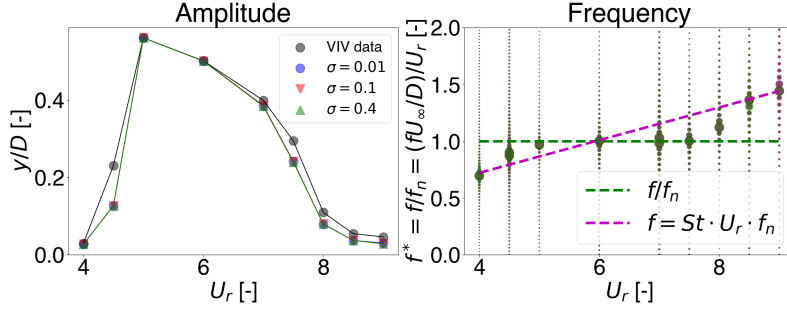


Fig. 6 Amplitude and Power Spectral Density (PSD) for displacement signal $y(\tau)$ for several reduced velocities. The typical lock-in scenario is recovered by the surrogate model.

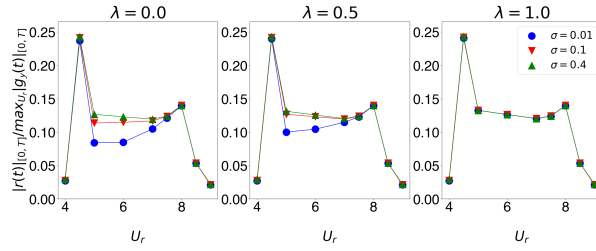


Fig. 7 Residual error (given as $|r(\tau)|_{[0,T]} \equiv \int_0^T |r(\tau)|^2 d\tau$, normalized by $\max_{U_r} |g_y(\tau)|_{[0,T]}$), function of the reduced velocity U_r for the finer regression model.

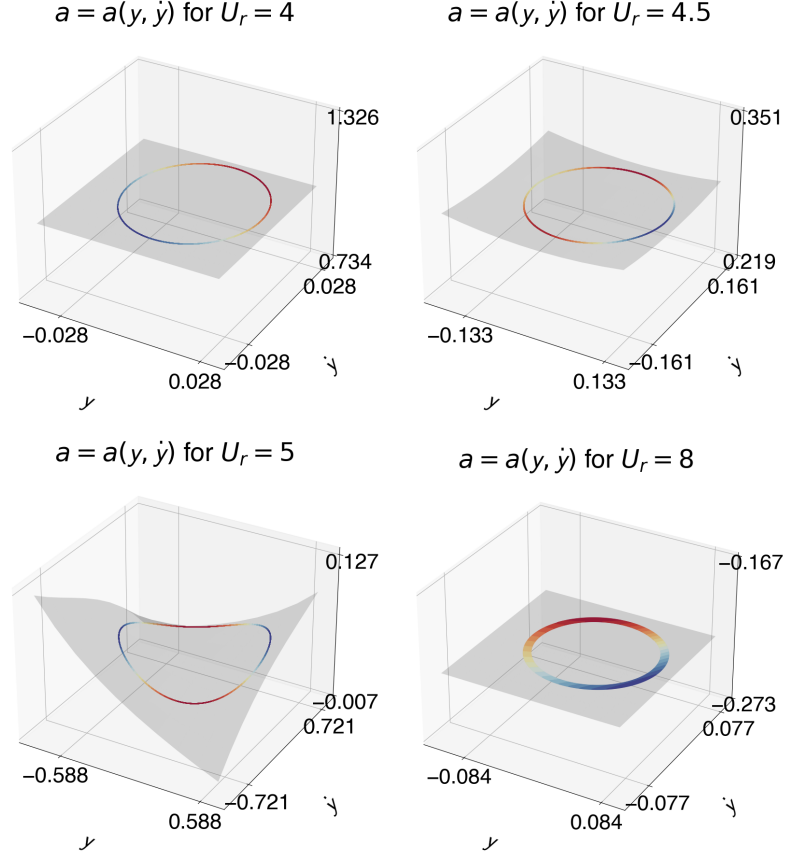


Fig. 8 $a(y, \dot{y})$ for $U_r = 4, 4.5, 5$ and 8 and for $\sigma = 0.4$ and $\lambda = 1$

dependency of a on $(y, \dot{y})$. The case $U_r = 5$ is the one that presents the most variation of a with $(y, \dot{y})$, showing that even if the signal seems, at first sight, sinusoidal, it is actually more complex, and can be modeled by this variable-mass framework.

In Figure 9, we present the simulation of the obtained model and compare it with the VIV data, in the same way we did the first regression case, presented in Figure 5. We see there is no substantial effect on the model's simulation. Also, we present in Figure 10 the added mass (both $a(\tau) = m_a(\tau)/m_c$ and $C_a = m_a(\tau)/m_f$), as function of time, for a few values of reduced velocity, all for $\lambda = 1$ and $\sigma = 0.4$. We can see that the major effect contributing to its change is the mean value, function of the reduced velocity. Its time variation, due to the varying state, as presented in Figure 9, is more important for $U_r = 5$ due to the higher complexity of the signal to be represented by the model, as discussed previously. In order to understand this behavior more clearly, we plot as well in Figure 10 the mean value (time-average) of $C_a(\tau)$ for all reduced velocities. We can see that it is very much comparable with the coefficient $\gamma_{0,0}$, related to the constant term in the model for $a = a(y, \dot{y})$. Moreover, we can also see the trends where, for reduced velocities before the lock-in, the added mass will be positive, zero

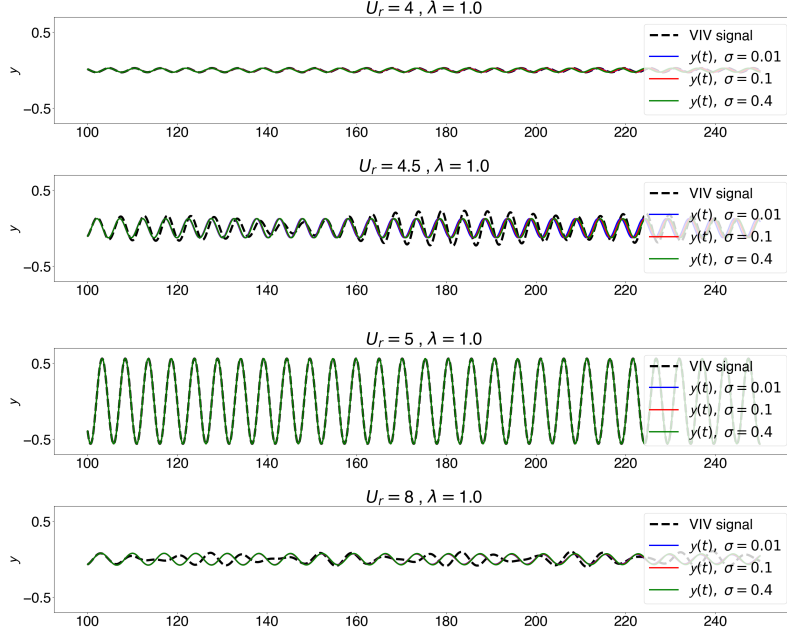


Fig. 9 Signal from cross-wise displacement ($y(\tau)$) for the model's outputs (in colors) and the comparison with original, raw signal, in black dashed lines, for $U_r = 4, 4.5, 5$ and 8 .

at the lock-in and it stays small (nearly zero, in fact) for a wide range and negative for post lock-in, as discussed before and documented in [6, 10, 32, 34]. As a final remark, we point out that the obtained value of a is always greater than -1 , for all reduced velocities. Such an important result ensures that the kinetic energy of the equivalent particle is always a positive quantity, as $m = m_c + m_a = m_c(1 + a) > 0$. This shows that the regression procedure produced a consistent physics-based surrogate model, without the need for additional constraints.

4 Conclusions

In this work, we employed the dynamics of variable mass systems formalism (as presented in [26]) to build a surrogate model for a rigid cylinder, mounted on an elastic-base, under Vortex-Induced Vibration (VIV). This model consists in a single equivalent particle whose mass is composed by the cylinder's one and a virtual one, the added-mass, which is let to depend on both the position and the velocity of that particle. This modeling choice is justifiable in order to capture memory effects of the viscous flow around the cylinder. We remark that time-dependent-only added masses were considered in [14], but such proposition could not lead to an autonomous surrogate model for the VIV phenomenon.

Through a data-driven regression process, we were able to identify such model and validate it, which presented good results for pre-lock-in, lock-in and post-lock-in regimes and adequate results for transitional regimes. For those transitional regimes,

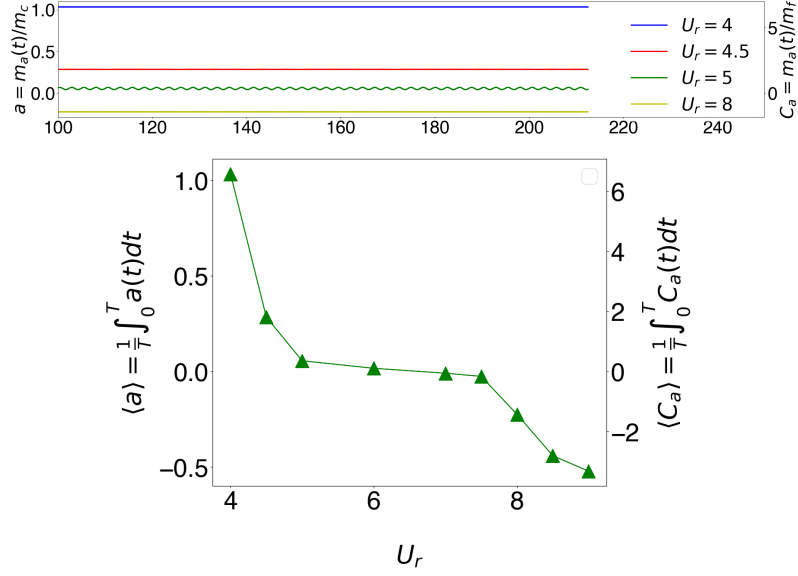


Fig. 10 $a = m_a(\tau)/m_c$ and $C_a = m_a(\tau)/m_f$, for $U_r = 4, 4.5, 5$ and 8 , and its time average for all U_r . Results given for $\sigma = 0.4$ and $\lambda = 1$. Notice that **the dimensionless added-mass is always greater than -1 , $a > -1$, (or equivalently $C_a > -m^*$).** For this reason, the mass of the equivalent particle $m = m_c + m_a = m_c(1 + a)$ will always be positive, leading to a proper form of the kinetic energy.

in which two non-commensurable frequencies occur in the signal, the design of an accurate single-degree-of-freedom model is a difficult task. A natural direction for future work would be the consideration of experimental data, for which a more realistic model could be designed. Fundamental cases are planned to be explored, as those related to forced oscillations and VIV, as in [31].

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The Mechanics of Variable Mass Systems applied to the Added Mass Concept of a Moving Cylinder in Water

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Abstract

This paper constructs a physics-based surrogate mathematical model for the motion of a cylinder surrounded by a fluid under a distinct point of view: the mechanics of variable mass systems. Particularly, the concept of added mass is addressed. First, we introduce a 1 DoF problem, consisting of an equivalent virtual particle, free to oscillate in one direction, whose mass is let to depend on its position and velocity. The resulting kinetic energy models that of the whole system, solid and surrounding fluid. A general formulation for the equation of motion is then proposed, by applying the Extended Lagrange Equations for variable mass systems, from which the surrogate model is derived. We take as first case study the classic Vortex-Induced Vibration (VIV) phenomenon of a cylinder mounted on an elastic base. Then, we assume the added mass as a polynomial function on position and velocity. The coefficients of this polynomial expansion are estimated by regression, where we minimize the residual between the model's response and external data, herein coming from Computational Fluid Dynamics (CFD) simulations. The results are rewarding and the well known behavior of the added mass as a function of the reduced velocity observed in the technical literature, from experiments and from CFD simulations, is consistently recovered.

Keywords: Extended Euler-Lagrange equations, variable mass systems, added mass, Vortex-Induced Vibrations, surrogate models

1 Introduction

Problems where variable-mass systems formulations are employed have been extensively studied in the past. A classical example would be the so-called "rocket problem", which burns and expels fuel over time, changing the mass within a control-volume corresponding to the carcass of the rocket. Another example, would be the classic falling chain problem (Cayley (1857)), in which the mass of the suspended part varies with the tip position, that may be taken as a generalized coordinate. More recently, the Euler-Lagrange equation of such variable-mass systems has been derived (see Pesce (2003)), allowing for a systematic way to establish the equations of motion, in cases where the mass depends not only explicitly on time, but also on position and velocity. For a review of such equations, written in a few different forms see Pesce et al. (2024), chapter 8. Many applications in classical mechanics have been studied through this lens (in addition to the falling chain, we may also point out the impact of solid body on a liquid surface, as in Pesce (2003) or Casetta et al. (2011); Casetta and Pesce (2006) for a more detailed description; the oscillating water column, see Silva et al. (2020); Da Silva et al. (2023) and Pesce et al. (2024), chapter 8); or even the dynamics of vertically collapsing buildings (see Pesce et al. (2012)), all of which had the mass dependent on some generalized position only. Although the extended Lagrange equation in Pesce (2003) has been derived for systems with mass explicitly dependent on time, position and velocity, the latter dependence has never been addressed in practical problems. Indeed, it is quite hard to conceive a mechanical system with mass depending explicitly on its velocity. However, an attempt to do so could be to use mass in a virtual manner, such as the hydrodynamical added mass concept in the context of viscous flows.

The general concept of added mass (see, for example, Lamb (1932)) aims at modeling the component of the force (generally reactive) exerted by a fluid on a moving solid, which is in anti-phase with its acceleration. In addition, this added mass is not dependent on the velocity of the solid, but just on its position, if in the proximity of another solid body or a free surface. We remark that, as the flow is ideal, the added mass can be seen as a geometrical concept, particular to the solid's shape and immersed volume (or drift volume). This point of view was explored in the work of Darwin (1953), and corresponds to the so-called Darwin's drift. Although firstly introduced (and fully justifiable) in the context of potential flow theory, the added mass concept can be extended empirically to model more complex scenarios in viscous flows, such as Vortex Induced Vibrations (VIV). In VIV, we typically witness the lock-in phenomenon, where the frequency of Vortex-Shedding (f_{VS} , or the external excitation frequency, from the point of view of the solid) matches the solid's natural frequency (f_n) and stays related to it when f_{VS} is increased, when augmenting the incoming flow velocity. This shift in frequency can be thus modeled as an added mass effect (see Sarpkaya (1977)), dependent on the frequency (or reduced velocity). This added mass is positive for frequencies f_{VS} below f_n (as f_{VS} dominates), zero for $f_{VS} \approx f_n$, negligible within the lock-in range, and negative for larger f_{VS} (see Fujarra (2002)). It turns out that this behavior is much more general, where any system, being forced with a well-defined-frequency forcing (see Cunha et al. (2006)) exhibit such behavior as well. More recently, a time-varying added mass (based on Darwin's drift, which allowed to

experimentally compute it), the so-called energized-mass, was introduced by [Galler \(2022\)](#) and allows for the modeling of viscous effects in time-evolving boundary layers ([Fernando et al. \(2017\)](#)). In this framework, the mass of the equivalent mechanical model varies with time, and the Euler-Lagrange equations in [Pesce \(2003\)](#) have been used in the derivation of the system's equations. However, although the energized-mass model can well reproduce the data, it does not provide a description based on a fully autonomous (whose mathematical structure is time-independent) mechanical system. To do so, its time-dependency would need to be replaced with the dependency on the system's state (position and velocity). This dependency would encode not only how the vortical pattern in the flow would force the moving solid, but would also have some memory effect (due to dependency on velocity), modeling viscous effects and boundary layer lags.

The present work can be divided in two main parts. First a system's dynamical equations are derived based on the Euler-Lagrange formalism presented in [Pesce \(2003\)](#), where the mass is allowed to be a general function of position and velocity. In this formulation, the mass that is added/lost by the particle has a certain velocity, and a discussion around how this quantity is modeled will be provided. Then, in a second part, the mass is allowed to be a polynomial function of position and velocity, whose coefficients are determined via regression on data coming from numerical simulations (Computational Fluid Dynamics, CFD). This step is similar to recent techniques such as SINDy (see [Brunton et al. \(2016\)](#) for its original proposition and [Honigbaum and Rochinha \(2022\)](#) in the context of VIV). We should also cite a more recent work from [Harduf et al. \(2024\)](#), who employed an algorithm to determine the system's inertia, also function of the state, based on the Hilbert Transform ([Lyons \(1997\)](#)), which, in a first step the frequency is determined as function of time, and further correlated to the state equation which also has a state-dependent natural frequency.

2 The variable mass system approach and the 1 DoF problem

First of all, let us introduce the general variable-mass model, which is intended to describe, as a surrogate model, the whole Vortex-Induced Vibration (VIV) problem by the virtue of the mass being explicitly dependent on the position, velocity and time [Pesce \(2003\)](#). The starting point will be the extended Lagrange Equation valid this type of systems:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j^a + Q_j^t + Q_j^y + Q_j^{\dot{q}} \quad (1)$$

with

$$\begin{aligned}
Q_j^a &= \sum_i \mathbf{f}_i \cdot \frac{\partial P_i}{\partial q_j} \\
Q_j^t &= \sum_i \dot{m}_i \mathbf{w}_i \cdot \frac{\partial P_i}{\partial q_j} \\
Q_j^q &= - \sum_i \left(\frac{1}{2} \frac{\partial m_i}{\partial q_j} |\mathbf{v}_i|^2 \right) \\
Q_j^{\dot{q}} &= \frac{d}{dt} \sum_i \left(\frac{1}{2} \frac{\partial m_i}{\partial \dot{q}_j} |\mathbf{v}_i|^2 \right)
\end{aligned} \tag{2}$$

In Eq. 2, $m = m(t, q_j, \dot{q}_j)$ is the mass of the j -th particle, explicitly depending on time, generalized coordinates and their time derivatives; $\mathbf{f}_i$ is an external force applied to the particle; $\mathbf{v}_i$ is its velocity and $\mathbf{w}_i$ is the velocity of the lost (gained) mass from (to) the particle, both measured w.r.t. the same inertial and cartesian frame of reference. Notice that the three latter terms in Eq. 2 are respectively functions of derivatives of the particle mass w.r.t. time, position, and velocity. For the case where $m = m(t, q_j)$, those equations are also presented in other forms, see [Pesce et al. \(2024\)](#), chapter 8.

Let us now particularize this model bring it closer to the VIV phenomenon. Let us consider a cylinder immersed in water, having the transversal oscillation (perpendicular to the incident flow) as the single degree of freedom. The kinetic energy of the system, composed by the cylinder and the surrounding fluid, will be modeled and summarized in a single equivalent particle:

$$T = \frac{1}{2} m(t, y, \dot{y}) \dot{y}^2 \tag{3}$$

Here, the mass of the equivalent particle $m(t, y, \dot{y})$ will be dependent on position, velocity and time. In general, this mass will be given by the sum of the cylinder's mass, m_c and an added mass contribution $m_a = m_a(t, y, \dot{y})$:

$$m(t, y, \dot{y}) = m_c + m_a(t, y, \dot{y}) \tag{4}$$

Then, the corresponding extended Lagrange Equation is written,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{y}} \right) - \frac{\partial T}{\partial y} = Q^a + Q^t + Q^y + Q^{\dot{y}} \tag{5}$$

with

$$\begin{aligned}
Q^a &= F(t, y, \dot{y}) = F_m + R \\
Q^t &= \dot{m}w \\
Q^y &= -\frac{1}{2} \frac{\partial m}{\partial y} \dot{y}^2 \\
Q^{\dot{y}} &= \frac{d}{dt} \left(\frac{1}{2} \frac{\partial m}{\partial \dot{y}} \dot{y}^2 \right)
\end{aligned} \tag{6}$$

In the term $F(t, y, \dot{y})$ we include all the other forces acting on the cylinder, which are not of inertial nature, written as a sum of F_m and R . The F_m term consists in the modeled forces, which are supposed to affect the dynamics, such as elastic or damping ones. In the term R we include the residual forces that could not be taken into account in the model F_m .

Here, w is the velocity of the lost (gained) mass w.r.t. the same inertial frame of reference used for measuring y (see [Pesce \(2003\)](#)). As a first ansatz, this velocity might be taken as null, $w = 0$, what is known as the Levi-Civita case in the mechanics of variable mass systems [Pesce \(2003\)](#). An alternative for that, would be to take $w = \dot{y}$, the same velocity of the studied particle, set equal to the body's. Such an alternative will be discussed further in this paper. For simplicity, we let the velocity of the gained or lost mass be a linear function of the equivalent particle's velocity, i.e.,

$$w(t) = \lambda \dot{y}(t) \tag{7}$$

where λ is taken as a constant. According to λ , we may have different cases:

- $\lambda = 0 \rightarrow w(t) = 0$: the particle acquires/loses mass at zero velocity. This is commonly known as the Levi-Civita case (see [Pesce \(2003\)](#))
- $\lambda = 1 \rightarrow w(t) = \dot{y}(t)$: the particle acquires/loses mass at the same velocity it moves
- $\lambda \notin \{0, 1\}$: the particle acquires/loses mass at some other velocity

We observe that this is a (not unique) choice that simplifies the problem. Other models could be considered, including additional transfer functions between the input ($\dot{y}$) and the output (w). This could be investigated in future work.

With this settings, the terms in the extended Euler-Lagrange equation are given by:

$$\frac{\partial T}{\partial y} = \frac{1}{2} \left(\frac{\partial m}{\partial y} \right) (\dot{y})^2, \quad \frac{\partial T}{\partial \dot{y}} = \frac{1}{2} \frac{\partial m}{\partial \dot{y}} (\dot{y})^2 + m \dot{y} \tag{8}$$

and

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{y}} \right) &= \frac{d}{dt} \left(\frac{1}{2} \frac{\partial m}{\partial \dot{y}} (\dot{y})^2 + m \dot{y} \right) \\
&= \frac{d}{dt} \left(\frac{1}{2} \frac{\partial m}{\partial \dot{y}} (\dot{y})^2 \right) + \left(\frac{\partial m}{\partial t} + \frac{\partial m}{\partial y} \dot{y} + \frac{\partial m}{\partial \dot{y}} \ddot{y} \right) \dot{y} + m \ddot{y}
\end{aligned} \tag{9}$$

We can see that, a few terms involving derivatives of the mass w.r.t. the state cancel out with Q^y and $Q^{\dot{y}}$, leading to:

$$\begin{aligned} \left(m_c + m_a + \frac{\partial m_a}{\partial \dot{y}}(1 - \lambda)\dot{y} \right) \ddot{y} + \left(\frac{\partial m_a}{\partial y} \right) \dot{y}^2(1 - \lambda) \\ + \left(\frac{\partial m_a}{\partial t} \right) \dot{y}(1 - \lambda) = F(t, y, \dot{y}) \end{aligned} \quad (10)$$

2.1 The equations of motion for a cylinder under VIV

So far, the equations of motion have been given in a general manner, for a single degree of freedom, variable mass system. We further particularize the model, taking a cylinder attached to a linear spring-damper system, of known constants k and c . Also, in order to model extra components of the fluid force on the solid, which were not necessarily captured by the (variable) added mass, we further add some typical non-linear viscous damping, whose force is given by $\tilde{\chi}|\dot{y}|\dot{y}$. The generalized force may be then written,

$$F_m(t, y, \dot{y}) = -c\dot{y} - ky - \tilde{\chi}|\dot{y}|\dot{y} \quad (11)$$

The surrogate model is finally given by:

$$\begin{aligned} \left(m_c + m_a + \frac{\partial m_a}{\partial \dot{y}}(1 - \lambda)\dot{y} \right) \ddot{y} + \left(\frac{\partial m_a}{\partial y} \right) \dot{y}^2(1 - \lambda) \\ + (c + \tilde{\chi}|\dot{y}|) \dot{y} + ky + \frac{\partial m_a}{\partial t}(1 - \lambda)\dot{y} = R \end{aligned} \quad (12)$$

Notice that $\lambda = 1$ leads to the simplest form for this surrogate model, making null all the terms coming from $\dot{m}_a = dm_a/dt$. This is going to be further explored in the paper, not only as the simplest form but also as a natural choice.

2.2 Non-dimensional model

Before going any further, we rewrite here the dimensionless form of the surrogate model, which will be convenient for further generic development. To do so, we give ourselves a typical length scale D (taken here as the cylinder's diameter) and a typical velocity scale U (to be precised later) and a resulting time-scale D/U , such that we can define:

$$\tau = \frac{t}{D/U}, \quad y = \frac{y}{D}, \quad \dot{y} \equiv \frac{d}{d\tau}y = \frac{\dot{y}}{U}, \quad \ddot{y} \equiv \frac{d^2}{d\tau^2}y = \frac{\ddot{y}D}{U^2} \quad (13)$$

where here we abuse the notation $(\dot{\cdot})$ for derivatives with respect to both t and τ . We may also define:

$$a \equiv \frac{m_a}{m_c} = a(\tau, y, \dot{y}), \quad (14)$$

to be the dimensionless added mass, function of $(\tau, y, \dot{y})$. Alternatively, as it is more usual in some studies, we may also make the added mass m_a dimensionless with the displaced volume mass, $m_d = \rho\pi D^2/4$:

$$C_a = \frac{m_a}{m_d} = a \cdot m^*, \quad \text{with} \quad m^* = \frac{m_c}{m_d}. \quad (15)$$

where ρ is the density of the surrounding fluid. Replacing those identities in equation 12, we get:

$$\begin{aligned} & \left(1 + a + \frac{\partial a}{\partial \dot{y}}(1 - \lambda)\dot{y}\right) \ddot{y} + \left(\frac{\partial a}{\partial y}\right) (\dot{y})^2(1 - \lambda) \\ & + \left(4\pi \frac{\zeta}{U_r} + \chi|\dot{y}|\right) \dot{y} + \left(\frac{2\pi}{U_r}\right)^2 y + \frac{\partial a}{\partial \tau}(1 - \lambda)\dot{y} = \frac{R}{m_c U^2/D} \equiv r \end{aligned} \quad (16)$$

with:

$$\zeta = \frac{1}{2} \frac{c}{\sqrt{k m_c}}, \quad f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m_c}}, \quad U_r = \frac{U}{D f_n}, \quad \chi = \frac{\tilde{\chi}}{m_c D} \quad (17)$$

the structural damping ratio ζ , the solid's natural frequency f_n (without the constant added mass correction, as it will be found by the data-driven regression process afterwards), the reduced-velocity U_r (based on f_n , as in Menon and Mittal (2021)) and the dimensionless nonlinear damping coefficient χ .

2.3 Model for the variable mass, $m_a = m_a(t, y, \dot{y})$ (or $a = a(\tau, y, \dot{y})$)

At this point, the added mass of the cylinder has been modeled, in all its generality, as function of time t , position y and velocity $\dot{y}$, $m_a = m(t, y, \dot{y})$, or, in its dimensionless form $a = a(\tau, y, \dot{y})$. The explicit time-dependency of the mass with time has been recently addressed by the introduction of the so-called energized mass concept Galler et al. (2021). Here, instead of letting it be explicitly function of time, we only let it be function of the position and velocity $(y, \dot{y})$, its time-dependency being implicitly given by the time variation of those variables. This latter view is more aligned with the philosophy of providing an autonomous description of the system.

We further suppose the added mass to be dependent on position and velocity as a polynomial:

$$a = a(y, \dot{y}) = \sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i,j} y^i \dot{y}^j \quad (18)$$

This is commonly done in system identification literature (see, for example Lontaris and Billings (1985); Aguirre and Letellier (2009); Brunton et al. (2016)) and is convenient when no better or more obvious functional basis is at hand. However, each

polynomial term has a physical interpretation. For instance, the first one, $\gamma_{0,0}$ is a time-invariant term and provides the classical "constant" added mass term, first introduced for potential flows and latter extended to the VIV phenomenon as function of the reduced velocity (see [Sarpkaya \(1979\)](#)). The terms involving the position-dependency only, i.e. $\gamma_{1,0}y + \gamma_{2,0}y^2 + \dots$ can also be interpreted as the portion of the added mass that models purely space-dependent masses, as in the case where nearby solid walls or free-surfaces are present. The remaining terms are responsible for modeling velocity-dependent masses, and thus for memory effects. With this model, the derivatives of the added mass w.r.t. y and $\dot{y}$ are given by:

$$\begin{aligned}\frac{\partial a}{\partial y} &= \sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i+1,j} (i+1) y^i \dot{y}^j \\ \frac{\partial a}{\partial \dot{y}} &= \sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i,j+1} (j+1) y^i \dot{y}^j\end{aligned}\tag{19}$$

We remark that even if the expression for the added mass a is polynomial in y and $\dot{y}$, the whole dynamics governing the state $(y, \dot{y})$ is not polynomial and has a rather complex structure. Indeed, the representation in state-space of this dynamical system would lead to a fractional function, whose numerator not only contains polynomials, but also a function of the type $|\dot{y}| \dot{y}$, if $\chi \neq 0$. In terms of model regression approaches, this does not correspond to a typical early versions of SINDy [Brunton et al. \(2016\)](#), although closer to more recent versions of it such as py-SINDy [Kaheman et al. \(2020\)](#). Notice that, in our case, this representation comes naturally from the variable-mass Lagrangian formalism [Pesce \(2003\)](#).

We also remark that, from a physical point of view, the only known parameters in the model are the mass of the cylinder, m_c , the linear damping c and the stiffness, k , leading to known values for ζ , f_n and U_r . All the other parameters, namely $\gamma_{i,j}$ and χ are "virtual" and will be determined by a regression step, as described in the next section. The parameter λ , which determines how the velocity of acquired mass will be related to the cylinder's velocity, will be assessed.

2.4 Reconstruction of added mass with regression of data

This section is devoted to the numerical procedure for determining the coefficients $\gamma_{i,j}, \chi$. We suppose we have at hand a time-series data for $y(t)$. In our case, this data will come from Computational Fluid Dynamics (CFD) simulations. However, experimental data could be used as well. With this data we proceed to compute its derivatives, up to second order $(\dot{y}, \ddot{y})(t)$, for instance using second-order finite difference scheme. The coefficients $\gamma_{i,j}$ of each monomial and χ will be found by performing a least-square regression, where we minimize the cost functional:

$$J(\gamma_{i,j}, \chi) = \frac{1}{2} \|r(\gamma_{i,j}, \chi)\|_2^2 + \frac{\sigma}{2} \left(\sum_{i=0}^{N_i} \sum_{j=0}^{N_j} \gamma_{i,j}^2 + \chi^2 \right) \quad (20)$$

being the second term a regularization component, penalizing equally the square of each coefficient $\gamma_{i,j}$ and χ . We emphasize here that, due to the adimensionalization performed before, all those terms in the equation are also dimensionless and can be compared with each other.

3 Application to Vortex-Induced Vibration Case

3.1 The numerical approach to the fluid-structure problem

As a starting point for our analysis, we consider the fluid-structure interaction phenomenon described before, but regarded as solution of the fluid flow equations, coupled with the solid dynamics. In what follows, we choose to directly present this problem in dimensionless form, following the same procedure presented for the surrogate model, where the mass, the length and velocity used, m_c, D, U are still the cylinder's mass and diameter and U will be chosen here as the velocity of the incoming flow (uniformly distributed far from the cylinder), respectively. With that, the cylinder's displacement y , will be modeled by:

$$\ddot{y} + 4\pi \frac{\zeta}{U_r} \dot{y} + \left(\frac{2\pi}{U_r} \right)^2 y = g_y \equiv \frac{4}{\pi} \frac{1}{m^*} f_y \quad (21)$$

where f_y is the force from the fluid flow on the direction of motion y , and can be computed as an integral over the solid wall Γ_s of the (dimensionless) stress tensor:

$$\underline{\sigma} = p\underline{I} - Re^{-1}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad (22)$$

where $Re = \rho U D / \mu$ is the Reynolds number, based on the typical fluid velocity U , the cylinder's diameter D , the fluid density ρ and viscosity μ . This tensor is function of the pressure field $p = p(\mathbf{x})$ and of the gradients of the velocity field $\mathbf{u} = (u_1, u_2) = \mathbf{u}(\mathbf{x})$ (also dimensionless), along the normal direction $\mathbf{n}$ and projected onto the versor aligned with y , $\mathbf{e}_2$ as:

$$f_y(\mathbf{u}, p) = \int_{\Gamma_s} \mathbf{e}_2 \cdot (p \cdot \underline{I} - \mu(\nabla_{\mathbf{x}} \mathbf{u} + \nabla_{\mathbf{x}} \mathbf{u}^T)) \mathbf{n} dS \quad (23)$$

Here, $\mathbf{x} = (x_1, x_2)$ is a point in the space where the fluid-flow is defined, and is supposed two-dimensional. The coordinates x_1 (with unit vector $\mathbf{e}_1$) and x_2 (with unit vector $\mathbf{e}_2$) correspond to the streamwise and crosswise directions.

The fluid motion, driven by an imposed far-field velocity aligned with $\mathbf{e}_1$ and normalized, $\mathbf{u}_\infty = \mathbf{e}_1$, is governed by the Navier-Stokes equations:

$$\partial_\tau \mathbf{u} + (\mathbf{u} - \mathbf{u}_m) \cdot \nabla_{\mathbf{x}} \mathbf{u} + \nabla_{\mathbf{x}} p = Re^{-1} \Delta_{\mathbf{x}} \mathbf{u}, \quad \nabla_{\mathbf{x}} \cdot \mathbf{u} = 0 \quad (24)$$

Here we make a choice of writing the Navier-Stokes equations in the Arbitrary Lagrangian Eulerian (ALE) form (see for example [Hirt et al. \(1997\)](#)), as it will be the

choice for our numerical implementation when taking the mesh-velocity, $\mathbf{u}_m$, as the cross-wise velocity of the cylinder, $\mathbf{u}_m = \mathbf{e}_2 \dot{y}$.

The Reynolds number will be kept constant and equal to $Re = 100$ in all our simulations (as in [Menon and Mittal \(2021\)](#)). The mass-ratio will also be kept constant and equal to $m^* \approx 6.4$ [Menon and Mittal \(2021\)](#). Also, no structural damping will be considered ($\zeta = 0$). The reduced velocity U_r will be varied within a range of values of $U_r \in [4, 9]$. We remark that by taking both the Reynolds and the reduced mass constants, we fix both the vortex-shedding frequency (characterized by a Strouhal number of $St = f_{VS}D/U_\infty \approx 0.18$, which is computed with a fix-cylinder CFD simulation) and the mass/diameter of the cylinder. For this reason, if we were to reproduce those settings experimentally, we would have to vary the stiffness of the spring on which the cylinder is mounted (instead of increasing the flow velocity, which is more usual in practice [Franzini et al. \(2008\)](#)).

In order to study this problem, we simulate it using the finite-element code FreeFEM++ [Hecht \(2012\)](#) for the spatial discretization of the fluid problem. A second-order backward finite difference for the time-derivative of the fluid problem [Karniadakis et al. \(1991\)](#) and a centered finite-difference in time is used for the solid equations.

3.2 Numerical Simulations data

The results of several time-stepping simulations, started with initial conditions corresponding to the steady-state (base-flow) solution added with a small perturbation along the unstable mode, coming from spectral analysis of the linearized Navier-Stokes equations (corresponding to the vortex-shedding mode multiplied by a small amplitude) can be appreciated in Figure 1 for $U_r = 4, 4.5, 5$ and 8 . We can see that we recover the usual behavior where for small reduced velocities, the amplitude of motion is small and it is caused by the forcing from the fluid, which oscillates at the vortex-shedding (VS) frequency, typically characterized by a Strouhal number $St = f_{VS}D/U_\infty \approx 0.18$. For $U_r \geq 5$, the cylinder's natural frequency matches the VS one, and a resonance occurs, increasing the amplitude of motion. This is the beginning of the lock-in region. If we further increase U_r , we still witness a large amplitude of motion and the frequency of that motion is locked to the cylinder's natural frequency, even if the VS one is not matching exactly that frequency. We see that the dimensionless frequency $f^* \equiv f/f_n = f(U_\infty/D)/U_r$ becomes unity in this region. For much larger reduced velocities, near $U_r = 8$, we begin to witness the end of this phenomenon, where the amplitude of the motion becomes again smaller and the frequencies in the signal slowly start to transition to the fluid one. A summary of this description is given in Figure 2 showing the amplitude of motion and the Power Spectral Density (PSD) of the signal (always computed for $\tau > 100$, the fully-converged portion) for each reduced velocity considered here. We remark that, although this results are very close from the one reported by [Menon and Mittal \(2021\)](#), obtained in a simplified, low-Reynolds number, two-dimensional setup, it is qualitatively different from more usual lock-in diagrams (for example, [Fujarra et al. \(2001\)](#); [Franzini et al. \(2008\)](#)), obtained experimentally, when higher Reynolds numbers are involved, three-dimensional effects and elasticity

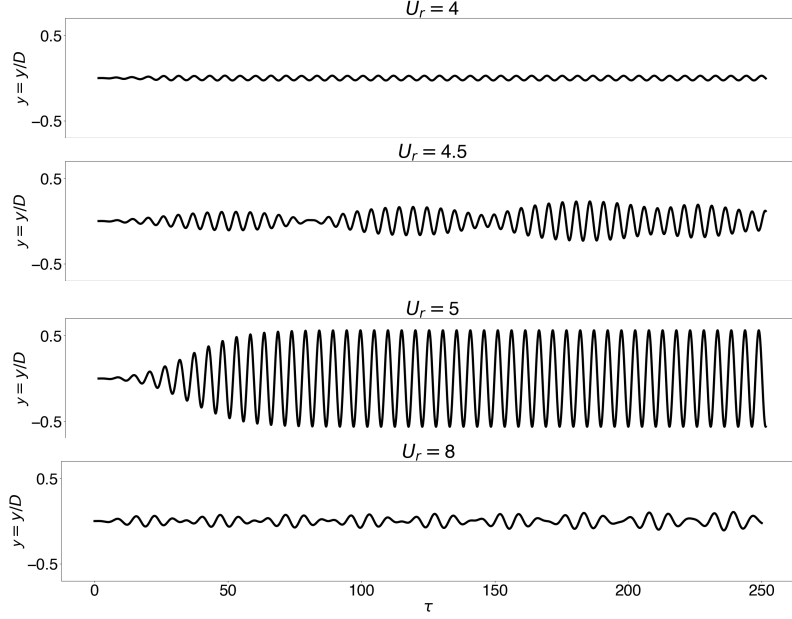


Fig. 1 Signal from cross-wise ($y(\tau)$) of the cylinder for $U_r = 4, 4.5, 5$ and 8 , determined from CFD simulations.

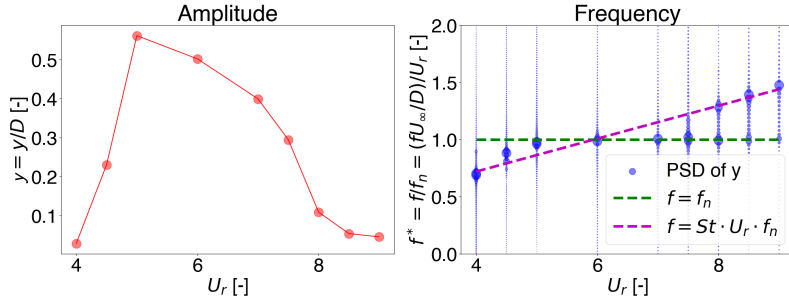


Fig. 2 Amplitude and Power Spectral Density (PSD, whose energy is proportional to the diameter of the circles) for displacement signal y for several reduced velocities. The typical lock-in scenario is recovered where higher-amplitudes are obtained for a reduced-velocity interval starting from $U_r \approx 5$, from which its frequency stops being correlated with the vortex-shedding one (characterized by a Strouhal number of $St \approx 0.18$) and correlates with the natural frequency of the solid $f = f_n$.

of the cylinder has to be taken into account. This simplification is employed for the sake of the methodology presentation and validation.

3.3 A first regression

In this section, we present some results in a first regression test, corresponding to polynomial degree of 2 in both y and $\dot{y}$, together with the regression in $\chi \dot{y} |\dot{y}|$. The

coefficients $\gamma_{i,j}$ for that regression are given in Figure 3, for three different regularization parameters $\sigma = 0.01, 0.1$ and 0.4 , and three different λ , which are $\lambda = 0$ (the mass is acquired at zero velocity, Levi-Civita case), $\lambda = 1$ (the mass is acquired at the same velocity as the particle) and an intermediate value of $\lambda = 0.5$. We can see that the first coefficient, $\gamma_{0,0}$ is nearly the same for all regularization parameters and for all acquisition velocity parameters. This means that, regardless of the physics of the added mass process and of the finer details of each regression, the constant term persists. Furthermore, we can interpret it as being the added mass coefficient, having a higher-value before the lock-in (where it has to adjust the system's natural frequency to match the one from the Vortex-shedding phenomenon) a zero-value within the lock-in (where the cylinder's natural frequency is already correct) and a negative value for post lock-in region (where again vortex-shedding dominates). The behavior for this coefficient had been reported for example in Vikestad et al. (2000); Fajarra (2002); Cunha et al. (2006) as experimental results and, for instance, in Willden and Graham (2001) by numerical simulations.

On top of this first adjustment of the system's natural frequency, we can observe the other polynomial coefficients for the model, $\gamma_{i,j}$. We can see that their behavior is typically active for reduced velocities where both solid and fluid frequencies are present (for example $U_r \approx 4.5$ and $U_r \approx 8$), showing that the surrogate model is trying to capture this more complex behavior. Interestingly, for the Levi-Civita case, all the model's coefficients exhibit a more intricate distribution at those regions, presenting higher values for $\gamma_{i,j}$. We can interpret this as coming from the fact that for $\lambda = 0$ (as opposed to $\lambda = 1$), the model has a more complex structure, exhibiting more nontrivial terms in its dynamics (equation 20), to be adjusted by the regression process. However, for $\lambda = 1$, the model becomes much simpler. This does not seem to be an issue for the regression error, given in Figure 4, which stays almost the same for all values of λ considered. As a matter of fact, $\lambda = 1$ seems a natural choice, considering the physical nature of the surrogate model, such that no 'continuous impacting force' is implied, as would be the case for $\lambda \neq 1$ (for instance, the Levi-Civita hypothesis, $\lambda = 0$), that seems unnatural. Furthermore, we can see that the error is practically the same for all regularization parameters σ as well. For this reason, in the following, we will take a closer look at the case $\lambda = 1$ and $\sigma = 0.4$, as they show to be the simplest ones.

For this case, we can see that the only active terms, besides $\gamma_{0,0}$ in the regression are $\gamma_{1,0}$, $\gamma_{2,0}$ and $\gamma_{0,2}$. The terms $\gamma_{1,0}$ and $\gamma_{2,0}$ represent a linear and a quadratic dependency on the position y . Having such dependency may seem counter-intuitive as it does not respect the y -symmetry of the VIV problem (i.e. if $(y(t), (u_1(x_1, x_2), u_2(x_1, x_2)), p(x_1, x_2))$ is a solution, so is $(-y(t), (u_1(x_1, -x_2), -u_2(x_1, -x_2)), p(x_1, -x_2)))$). Indeed, although the mass of the system has a quadratic dependency on y (which is symmetric w.r.t. negative/positive values of y), it also has a linear dependency on y , implying that different masses would be obtained for negative or positive values of y . From the point of view of potential flow (for which the added mass concept has first been introduced), the added mass should only depend on the position if, for example, solid walls near the cylinder are present, which would make the flow's streamlines to deform and affect differently the cylinder according to its position. However, we must have in mind that, according to

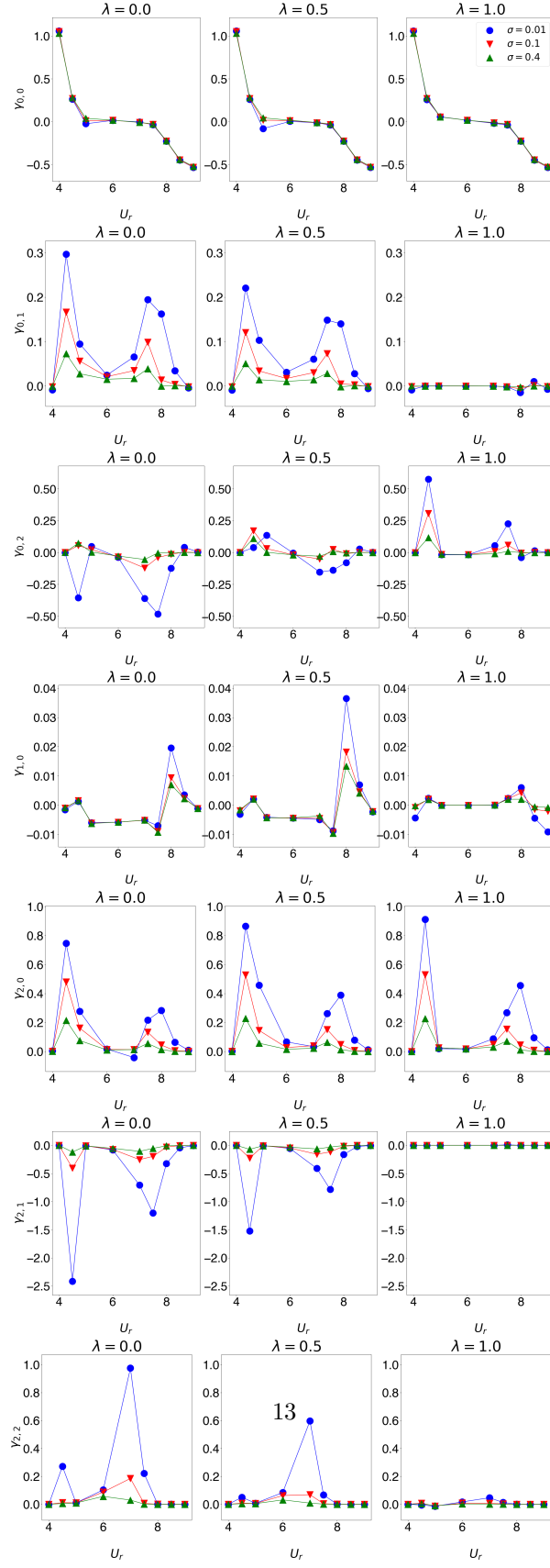


Fig. 3 Regression coefficients for the added mass, $\gamma_{i,j}$, as function of the reduced velocity U_r .

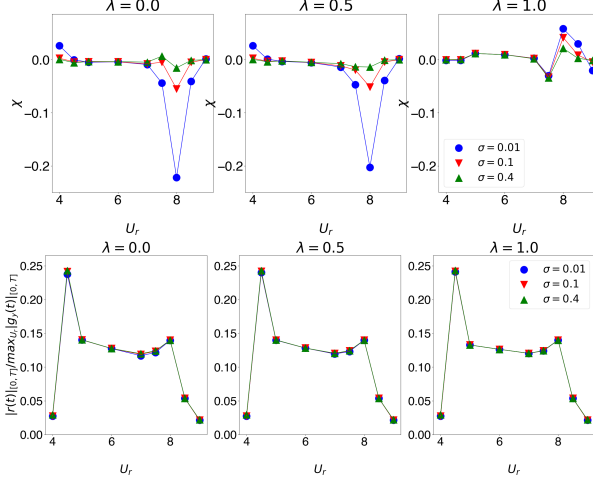


Fig. 4 Regression coefficient χ and residual error (given as $|r(\tau)|_{[0,T]} \equiv \int_0^T |r(\tau)|^2 dt$, normalized by $\max_{U_r} |g_y(\tau)|_{[0,T]}$), both function of the reduced velocity U_r .

the position of the cylinder, $y(\tau)$ (differently from the potential-flow case), the flow-field will exhibit a pattern corresponding to a given phase of the Vortex-Shedding motion. This pattern will affect the solid body in a different manner, according to the Biot-Savart law, translating the vortices in the fluidflow into a force, whose effect is modeled by the added mass. The term $\gamma_{0,2}$ represents a quadratic dependency in the velocity y for the added mass m_a , which is positive in all scenarios. This means that, if the cylinder is moving faster, its effective mass is higher. We remark that, for $\lambda = 1$ and $\sigma = 0.4$, this effect will mostly be present for $U_r \approx 4.5$ and $U_r \approx 8$, transitional regimes which present a more complex behavior.

Lastly, we should test the surrogate model and compare it with the actual signal coming from the original CFD results. The model has been simulated using an Ordinary Differential Equation (ODE) from "scipy". The initial condition for those simulations correspond to the actual state of the cylinder at $\tau = 100$ and are integrated on the same time grid up to $\tau = 250$. The results are provided in Figure 5.

As pointed out previously, due to an increased complexity of the signal, for $U_r = 4$ and $U_r = 5$, the model, for the three regularization parameters, perform very well. For the case $U_r = 4.5$, even with a complex added mass dependency on $(y, \dot{y})$, the model behaves essentially as a linear oscillator, not being able to fully exhibit the modulation presented before. Also, the model for $U_r = 8$ is not fully able to well represent the dynamics of the two distinct frequencies, as it seems only one is effectively present in the signal. This is also verified by Figure 6, where only one frequency is recovered by the PSD, whereas two distinct values are present in the original signal in Figure 2. Despite those deficiencies, we can see that the overall amplitude produced by the model captures well the actual one, showing that the model (although simple) can be used as first-order approximation of the dynamics.

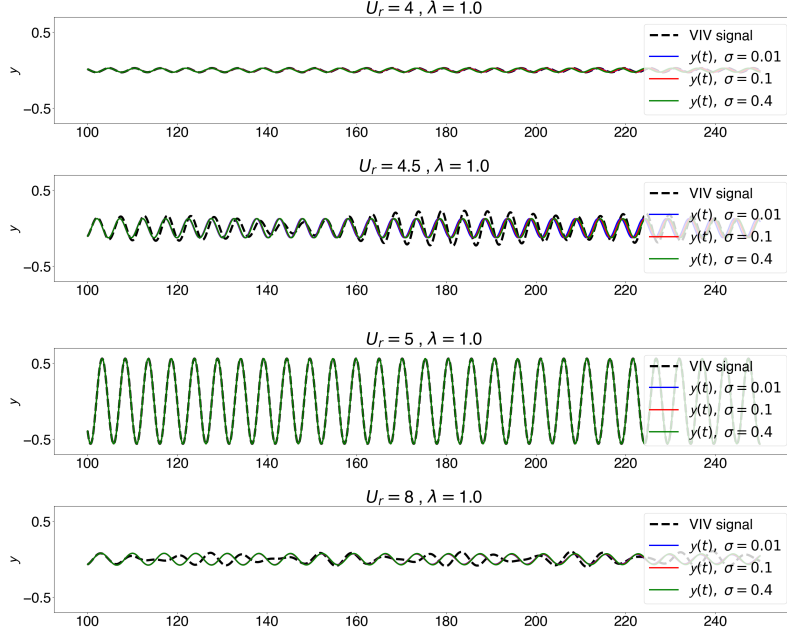


Fig. 5 Signal from cross-wise displacement (y) for the model's outputs (in colors; given by equation 20) and the comparison with original, VIV signal, in black dashed lines, for $U_r = 4, 4.5, 5$ and 8 .

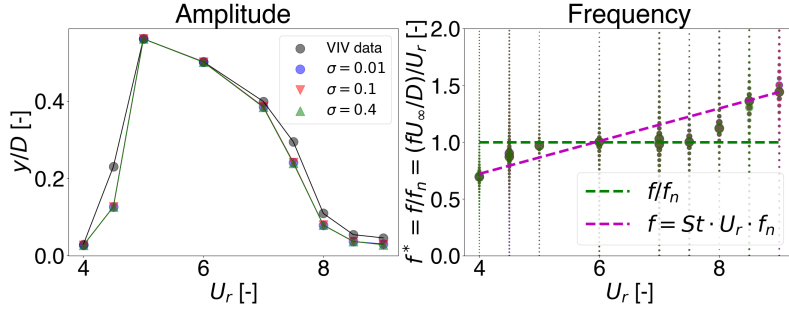


Fig. 6 Amplitude and Power Spectral Density (PSD) for displacement signal $y(\tau)$ for several reduced velocities. The typical lock-in scenario is recovered by the surrogate model.

3.4 A finer regression

The question that arises now is whether taking a higher polynomial order for the model can mitigate some of the deficiencies of the regressed model. To do so, we take now a fourth-order polynomial in both y and $\dot{y}$ variables. Instead of showing the coefficients, it is clearer and more concise to only show the error committed by the model, as presented in figure 7. We can see that by taking a higher-order model can only slightly mitigate some of the error especially for the less regularized case, $\sigma = 0.01$ and also for $\lambda = 0$. This is due to a richer structure of the surrogate model, as discussed before. Furthermore, this slight reduction in the error occurs in the lock-in region, when the

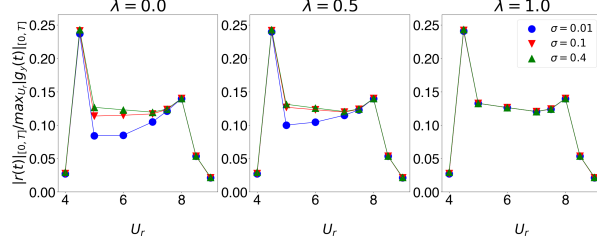


Fig. 7 Residual error (given as $|r(\tau)|_{[0,T]} \equiv \int_0^T |r(\tau)|^2 d\tau$, normalized by $\max_{U_r} |g_y(\tau)|_{[0,T]}$), function of the reduced velocity U_r for the finer regression model.

signal is rather simple, having essentially the natural frequency of the structure. This means that the essential of the dynamics has been captured with the previous model.

For the present model, also for the most regularized case, $\sigma = 0.4$, we present the added mass, for a few values of reduced velocity, as function of y and $\dot{y}$ in Figure 8. We can see that, for $U_r = 4$, $a = m_a/m_c$ is almost constant and equal to the added mass coefficient of nearly $a \approx 1$. For $U_r = 8$, a is also relatively constant, having $\gamma_{0,0}$ as its main contribution, but negative this time. For $U_r = 4.5$ we can see a slightly stronger dependency of a on $(y, \dot{y})$. The case $U_r = 5$ is the one that presents the most variation of a with $(y, \dot{y})$, showing that even if the signal seems, at first sight, sinusoidal, it is actually more complex, and can be modeled by this variable-mass framework.

In Figure 9, we present the simulation of the obtained model and compare it with the VIV data, in the same way we did the first regression case, presented in Figure 5. We see there is no substantial effect on the model's simulation. Also, we present in Figure 10 the added mass (both $a(\tau) = m_a(\tau)/m_c$ and $C_a = m_a(\tau)/m_f$), as function of time, for a few values of reduced velocity, all for $\lambda = 1$ and $\sigma = 0.4$. We can see that the major effect contributing to its change is the mean value, function of the reduced velocity. Its time variation, due to the varying state, as presented in figure 9, is more important for $U_r = 5$ due to the higher complexity of the signal to be represented by the model, as discussed previously. In order to understand this behavior more clearly, we plot as well in Figure 10 the mean value (time-average) of $C_a(\tau)$ for all reduced velocities. We can see that it is very much comparable with the coefficient $\gamma_{0,0}$, related to the constant term in the model for $a = a(y, \dot{y})$. Moreover, we can also see the trends where, for reduced velocities before the lock-in, the added mass will be positive, zero at the lock-in and it stays small (nearly zero, in fact) for a wide range and negative for post lock-in, as discussed before and documented in [Fujarra \(2002\)](#); [Cunha et al. \(2006\)](#); [Vikestad et al. \(1997\)](#); [Willden and Graham \(2001\)](#). As a final remark, we point out that the obtained value of a is always greater than -1 , for all reduced velocities. Such an important result ensures that the kinetic energy of the equivalent particle is always a positive quantity, as $m = m_c + m_a = m_c(1 + a) > 0$. This shows that the regression procedure produced a consistent physics-based surrogate model, without the need for additional constraints.

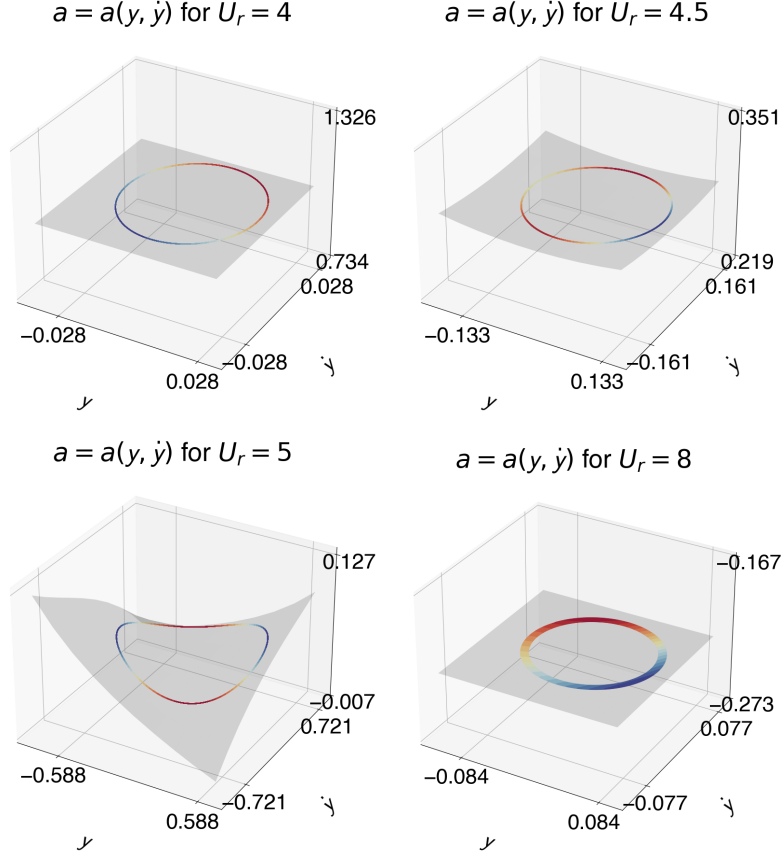


Fig. 8 $a(y, \dot{y})$ for $U_r = 4, 4.5, 5$ and 8 and for $\sigma = 0.4$ and $\lambda = 1$

4 Conclusions

In this work, we employed the dynamics of variable mass systems formalism (as presented in [Pesce \(2003\)](#)) to the Vortex-Induced Vibration (VIV) phenomenon, where a solid body, consisting in a cylinder, has its mass, sum of the physical mass and the added mass, dependent on both position and velocity. We remark that the added mass, being of virtual nature, may have its value dependent on time, position and velocity. This modeling choice is justifiable in order to capture memory effects of the viscous flow around the cylinder, creating a single-degree-of-freedom surrogate model for the VIV phenomenon. We remark that time-dependent-only added masses were considered in [Galler \(2022\)](#), but such proposition could not lead to an autonomous surrogate model for the VIV phenomenon.

Through a data-driven regression process, we were able to identify such model and validate it, which presented good results for pre-lock-in, lock-in and post-lock-in regimes and adequate results for transitional regimes. For those transitional regimes, in which two non-commensurable frequencies occur in the signal, the design of an

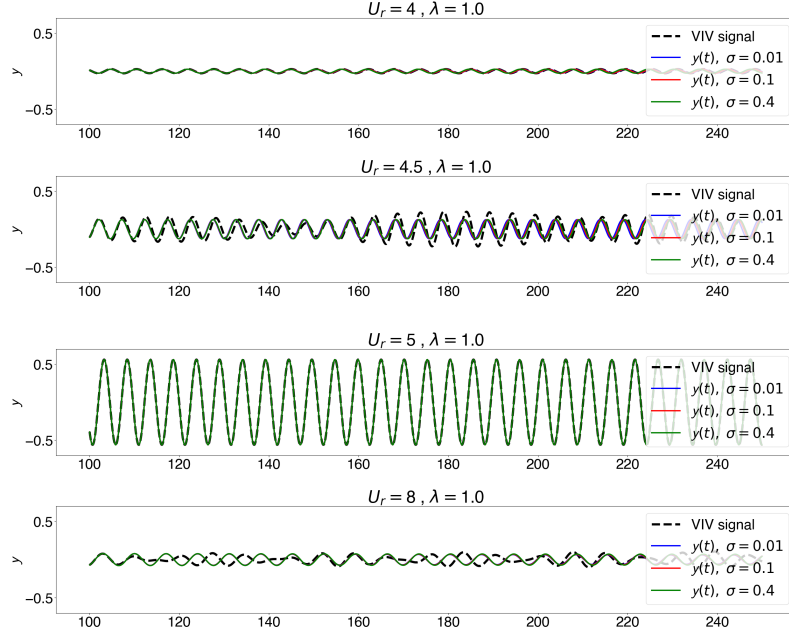


Fig. 9 Signal from cross-wise displacement ($y(\tau)$) for the model's outputs (in colors) and the comparison with original, raw signal, in black dashed lines, for $U_r = 4, 4.5, 5$ and 8 .

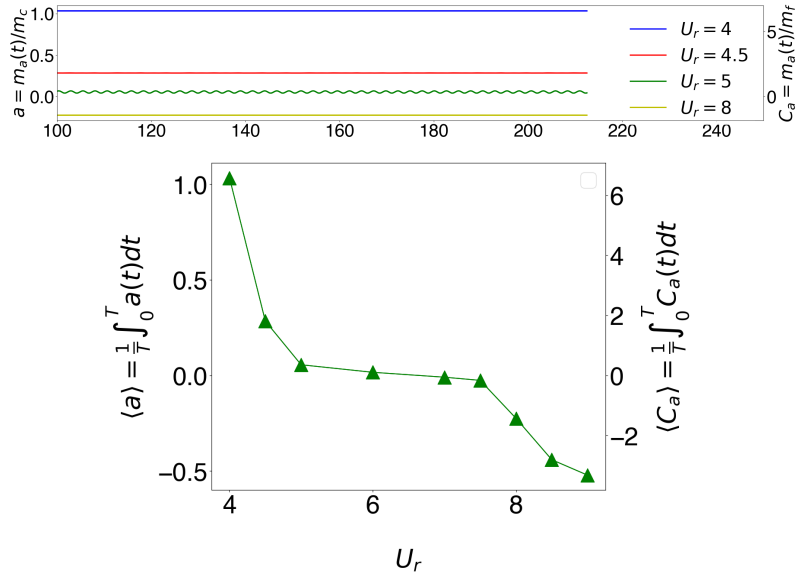


Fig. 10 $a = m_a(\tau)/m_c$ and $C_a = m_a(\tau)/m_f$, for $U_r = 4, 4.5, 5$ and 8 , and its time average for all U_r . Results given for $\sigma = 0.4$ and $\lambda = 1$. Notice that $a > -1$ ($C_a > -m^*$), result that is consistent with kinetic energy of the equivalent particle be always a non-negative quantity.

accurate single-degree-of-freedom model is a difficult task. A natural direction for future work would be the consideration of experimental data, for which a more realistic model could be designed. Fundamental cases are planned to be explored, as those related to forced oscillations and VIV, as in [Sumer and Fredsøe \(1988\)](#).

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